

Class 11 Computer Science (Python)

Complete Practical Programs (With Output & Clean Variable Names)

1 PRIME NUMBER

```
number = int(input("Enter a number: "))
is_prime = True

if number <= 1:
    is_prime = False
else:
    for divisor in range(2, int(number ** 0.5) + 1):
        if number % divisor == 0:
            is_prime = False
            break

if is_prime:
    print(f"{number} is a prime number.")
else:
    print(f"{number} is not a prime number.")
```

Sample Output

```
Enter a number: 7
7 is a prime number.
```

2 REVERSE A NUMBER

```
number = int(input("Enter a number: "))
reversed_number = 0

while number > 0:
    last_digit = number % 10
    reversed_number = reversed_number * 10 + last_digit
    number //= 10

print("Reversed number:", reversed_number)
```

Sample Output

```
Enter a number: 1234
Reversed number: 4321
```



3 PERFECT NUMBER

```
number = int(input("Enter a number: "))
divisor_sum = 0

for divisor in range(1, number):
    if number % divisor == 0:
        divisor_sum += divisor

if divisor_sum == number:
    print(f"{number} is a perfect number.")
else:
    print(f"{number} is not a perfect number.")
```

Sample Output

```
Enter a number: 6
6 is a perfect number.
```



4 ARMSTRONG NUMBER

```
number = int(input("Enter a number: "))
digit_count = len(str(number))
power_sum = 0

for digit in str(number):
    power_sum += int(digit) ** digit_count

if power_sum == number:
    print(f"{number} is an Armstrong number.")
else:
    print(f"{number} is not an Armstrong number.")
```

Sample Output

```
Enter a number: 153
153 is an Armstrong number.
```



5 PALINDROME NUMBER

```
number = int(input("Enter a number: "))

if str(number) == str(number)[::-1]:
    print("Palindrome")
else:
    print("Not Palindrome")
```

Sample Output

```
Enter a number: 121
Palindrome
```

**6**

FACTORIAL

```
number = int(input("Enter a number: "))
factorial_value = 1

if number < 0:
    print("Invalid number")
else:
    for i in range(1, number + 1):
        factorial_value *= i

    print("Factorial:", factorial_value)
```

Sample Output

```
Enter a number: 5
Factorial: 120
```

**7**

FIBONACCI SERIES

```
terms = int(input("Enter number of terms: "))
first, second = 0, 1

for _ in range(terms):
    print(first, end=" ")
    first, second = second, first + second
```

Sample Output

```
Enter number of terms: 5
0 1 1 2 3
```

**8**

HCF & LCM

```
num1 = int(input("Enter first number: "))
num2 = int(input("Enter second number: "))

hcf_value = 1
for divisor in range(1, min(num1, num2) + 1):
    if num1 % divisor == 0 and num2 % divisor == 0:
        hcf_value = divisor

lcm_value = max(num1, num2)
while True:
    if lcm_value % num1 == 0 and lcm_value % num2 == 0:
        break
    lcm_value += 1

print("HCF:", hcf_value)
print("LCM:", lcm_value)
```

Sample Output

```
Enter first number: 12
Enter second number: 18
HCF: 6
LCM: 36
```

**9**

STRING PALINDROME

```
text = input("Enter a string: ")

if text == text[::-1]:
    print("Palindrome")
else:
    print("Not Palindrome")
```

Sample Output

```
Enter a string: Krishna
Not Palindrome
```

**10**

VOWEL COUNT

```
text = input("Enter a string: ")
vowels = "aeiou"
vowel_count = 0

for char in text:
    if char.lower() in vowels:
        vowel_count += 1

print("Number of vowels:", vowel_count)
```

Sample Output

```
Enter a string: Krishna Anand
Number of vowels: 4
```

1 1 CHARACTER FREQUENCY

```
text = input("Enter a string: ")
char_frequency = {}

for char in text:
    char_frequency[char] = char_frequency.get(char, 0) + 1

print(char_frequency)
```

Sample Output

```
Enter a string: Krishna
{'K': 1, 'r': 1, 'i': 1, 's': 1, 'h': 1, 'n': 1, 'a': 1}
```

1 2 LIST FUNCTIONS DEMO

```
# List Functions Usage Demonstration

numbers = [10, 20, 30, 40, 20, 50]
print("Original List:", numbers)

# Adding elements
numbers.append(60)
numbers.insert(2, 25)
numbers.extend([70, 80])
print("After Adding Elements:", numbers)

# Removing elements
```

```

numbers.remove(20) # removes first occurrence
removed_last = numbers.pop()
removed_index = numbers.pop(1)
print("After Removing Elements:", numbers)
print("Removed Last:", removed_last)
print("Removed at Index 1:", removed_index)

# Searching
print("Index of 30:", numbers.index(30))
print("Count of 20:", numbers.count(20))

# Sorting & Reversing
numbers.sort()
print("Sorted List:", numbers)

numbers.reverse()
print("Reversed List:", numbers)

# Copying & Clearing
copied_list = numbers.copy()
copied_list.clear()
print("Cleared Copy:", copied_list)

# Built-in operations
print("Length:", len(numbers))
print("Maximum:", max(numbers))
print("Minimum:", min(numbers))
print("Sum:", sum(numbers))

# Slicing
print("Slice [1:4]:", numbers[1:4])
print("Slice [::2]:", numbers[::2])

```

Sample Output

```

Original List: [10, 20, 30, 40, 20, 50]
After Adding Elements: [10, 20, 25, 30, 40, 20, 50, 60, 70, 80]
After Removing Elements: [10, 30, 40, 20, 50, 60, 70]
Removed Last: 80
Removed at Index 1: 25
Index of 30: 1
Count of 20: 1
Sorted List: [10, 20, 30, 40, 50, 60, 70]
Reversed List: [70, 60, 50, 40, 30, 20, 10]
Cleared Copy: []
Length: 7
Maximum: 70
Minimum: 10
Sum: 280
Slice [1:4]: [60, 50, 40]
Slice [::2]: [70, 50, 30, 10]

```



1 3 TUPLE FUNCTIONS DEMO

```
values = (10, 20, 30, 40, 20, 50)

print("Count of 20:", values.count(20))
print("Index of 30:", values.index(30))
print("Max:", max(values))
print("Min:", min(values))
print("Sum:", sum(values))
```

📄 Sample Output

```
Count of 20: 2
Index of 30: 2
Max: 50
Min: 10
Sum: 170
```



1 4 DICTIONARY FUNCTIONS DEMO

```
# Dictionary Functions Usage Demonstration

student = {
    "name": "Krishna",
    "age": 16,
    "class": 11
}

print("Original Dictionary:", student)

# Accessing values
print("student['name']:", student["name"])
print("get('age'):", student.get("age"))

# Adding / Updating
student["school"] = "KV"
student.update({"age": 17})
print("After Adding & Updating:", student)

# Removing items
removed_class = student.pop("class")
print("After pop('class'):", student)
print("Removed Value:", removed_class)

removed_last = student.popitem()
print("After popitem():", student)
```

```

print("Removed Item:", removed_last)

# Dictionary views
print("Keys:", student.keys())
print("Values:", student.values())
print("Items:", student.items())

# Copying
copied_student = student.copy()
print("Copied Dictionary:", copied_student)

# Creating new dictionary
new_dict = dict.fromkeys(["a", "b", "c"], 0)
print("fromkeys():", new_dict)

# Length
print("Length:", len(student))

# Looping
print("Looping through dictionary:")
for key in student:
    print(key, ":", student[key])

```

Sample Output

```

Original Dictionary: {'name': 'Krishna', 'age': 16, 'class': 11}
student['name']: Krishna
get('age'): 16
After Adding & Updating: {'name': 'Krishna', 'age': 17, 'class': 11, 'school': 'KV'}
After pop('class'): {'name': 'Krishna', 'age': 17, 'school': 'KV'}
Removed Value: 11
After popitem(): {'name': 'Krishna', 'age': 17}
Removed Item: ('school', 'KV')
Keys: dict_keys(['name', 'age'])
Values: dict_values(['Krishna', 17])
Items: dict_items([('name', 'Krishna'), ('age', 17)])
Copied Dictionary: {'name': 'Krishna', 'age': 17}
fromkeys(): {'a': 0, 'b': 0, 'c': 0}
Length: 2
Looping through dictionary:
name : Krishna
age : 17

```



1

5

STRING FUNCTIONS DEMO

```

text = " Hello World 123 python STRING test "
print("Original String:", repr(text))

```



```

# Case Conversion
print("lower():", text.lower())
print("upper():", text.upper())
print("title():", text.title())
print("capitalize():", text.capitalize())
print("swapcase():", text.swapcase())
print("casefold():", text.casefold())

# Searching
print("find('World'):", text.find("World"))
print("rfind('o'):", text.rfind("o"))
print("index('Hello'):", text.index("Hello"))
print("count('o'):", text.count("o"))
print("startswith(' He'):", text.startswith(" He"))
print("endswith('test '):", text.endswith("test "))

# Testing
print("isalnum():", text.isalnum())
print("isalpha():", text.isalpha())
print("isdigit():", text.isdigit())
print("islower():", text.islower())
print("isupper():", text.isupper())
print("isspace():", text.isspace())
print("istitle():", text.istitle())

# Modifying
print("strip():", text.strip())
print("lstrip():", text.lstrip())
print("rstrip():", text.rstrip())
print("replace('python','Java'):", text.replace("python", "Java"))
print("split():", text.split())
print("rsplit():", text.rsplit())
print("split('o'):", text.split("o"))
print("partition('World'):", text.partition("World"))
print("rpartition('o'):", text.rpartition("o"))

# Joining
letters = ['A', 'B', 'C']
print("'-''.join(letters):", "-".join(letters))

# Encoding
print("encode():", text.encode())

```

Sample Output

```

Original String: ' Hello World 123 python STRING test '
lower():  hello world 123 python string test
upper():  HELLO WORLD 123 PYTHON STRING TEST
title():  Hello World 123 Python String Test
capitalize():  hello world 123 python string test
swapcase():  hELLO wORLD 123 PYTHON string TEST

```

```

casefold():  hello world 123 python string test
find('World'): 8
rfind('o'): 22
index('Hello'): 2
count('o'): 3
startswith(' He'): True
endswith('test '): True
isalnum(): False
isalpha(): False
isdigit(): False
islower(): False
isupper(): False
isspace(): False
istitle(): False
strip(): Hello World 123 python STRING test
rstrip(): Hello World 123 python STRING test
lstrip(): Hello World 123 python STRING test
replace('python','Java'):  Hello World 123 Java STRING test
split(): ['Hello', 'World', '123', 'python', 'STRING', 'test']
rsplit(): ['Hello', 'World', '123', 'python', 'STRING', 'test']
split('o'): [' Hell', ' W', 'rld 123 pyth', 'n STRING test ']
partition('World'): (' Hello ', 'World', ' 123 python STRING test ')
rpartition('o'): (' Hello World 123 pyth', 'o', 'n STRING test ')
'-'.join(letters): A-B-C
encode(): b' Hello World 123 python STRING test '

```


1
6

STAR PATTERN

```

rows = int(input("Enter number of rows: "))
pattern_type = input("Enter 'asc' or 'des': ").lower()

if pattern_type == "asc":
    for i in range(1, rows + 1):
        print("*" * i)

elif pattern_type == "des":
    for i in range(rows, 0, -1):
        print("*" * i)

else:
    print("Invalid choice")

```

Sample Output

```

Enter number of rows: 3
Enter 'asc' or 'des': asc
*

```

```
**  
***
```

1 7 SWAP TWO NUMBERS

```
a = float(input("Enter a: "))  
b = float(input("Enter b: "))  
  
a, b = b, a  
  
print("After swap:")  
print("a =", a)  
print("b =", b)
```

Sample Output

```
Enter a: 12  
Enter b: 95  
a = 95  
b = 12
```

1 8 SERIES SUM (AP & GP)

```
# AP Sum  
first_term = float(input("Enter first term (a): "))  
common_difference = float(input("Enter common difference (d): "))  
terms = int(input("Enter number of terms (n): "))  
  
ap_sum = terms * (2 * first_term + (terms - 1) * common_difference) / 2  
print("Sum of AP:", ap_sum)  
  
# GP Sum  
first_term = float(input("Enter first term (a): "))  
common_ratio = float(input("Enter common ratio (r): "))  
terms = int(input("Enter number of terms (n): "))  
  
if common_ratio == 1:  
    gp_sum = first_term * terms  
else:  
    gp_sum = first_term * (1 - common_ratio ** terms) / (1 - common_ratio)  
  
print("Sum of GP:", gp_sum)
```

Sample Output

```
Enter first term (a): 2
Enter common difference (d): 3
Enter number of terms (n): 7
Sum of AP: 77.0
Enter first term (a): 2
Enter common ratio (r): 6
Enter number of terms (n): 7
Sum of GP: 111974.0
```



1 9 RANDOM MODULE USAGE

```
import random

print("Random float:", random.random())
print("Random integer (1-10):", random.randint(1, 10))

numbers = [10, 20, 30, 40, 50]
print("Random choice:", random.choice(numbers))

random.shuffle(numbers)
print("Shuffled list:", numbers)

print("Random sample of 2:", random.sample(numbers, 2))
```

Sample Output

```
Random float: 0.1856777852708128
Random integer (1-10): 8
Random choice: 20
Shuffled list: [10, 40, 30, 50, 20]
Random sample of 2: [50, 20]
```



2 0 MATH MODULE USAGE

```
import math

print("Square root of 25:", math.sqrt(25))
print("2 power 5:", math.pow(2, 5))
print("Ceiling 4.3:", math.ceil(4.3))
print("Floor 4.7:", math.floor(4.7))
print("Factorial of 5:", math.factorial(5))
print("Value of pi:", math.pi)
print("GCD of 12 and 18:", math.gcd(12, 18))
```

Sample Output

```
Square root of 25: 5.0
2 power 5: 32.0
Ceiling 4.3: 5
Floor 4.7: 4
Factorial of 5: 120
Value of pi: 3.141592653589793
GCD of 12 and 18: 6
```

Best of Luck!

Write clean code. Avoid using variable names like `sum` , `list` , `str` . Maintain proper indentation. Stay calm.
You're prepared 🤖🔥